

Results — reading the April 2026 run, territory by territory

This note walks through the six polygons of the four case studies as they appear in the **April 2026 run outputs** (MapBiomass Collection 9, 1985–2023; Hansen GFC to 2024), archived under [~/Documents/2026_abc/four_indigenous_territories/data/yvynation_reflex_results/](#). Numbers are taken from the exported [comparison_1985_vs_2023.csv](#), [hansen_gfc_summary.csv](#), and [hansen_gfc_loss_by_year.csv](#) in each folder, and cross-referenced to the chapter tables. See [METHODOLOGY.md](#) for how these were produced and [SYNTHESIS.md](#) for the cross-case argument.

The four cases are ordered **coast** → **west**, which is also **old** → **recent** in the timing of their critical juncture. Read that way, the sequence is a single argument: *the later the protective policy arrived relative to the frontier, the more forest it was able to keep*.

Folder	Polygon	Extent	Headline 1985 → 2023 story
Krenak_23401_...1504	Krenak	4,282 ha	Forest +141% on former pasture — recovery
Krenak_de_Setee_Saloos_23402_...1512	Krenak Setee Salões	16,596 ha	Pasture –24%, Forest +6% — quiet mosaic shift
Bacurizinho_4901_...1148	Bacurizinho (1983 demarcated)	82,515 ha	Forest –3.6%, new pasture/soy/cane at edges
Bacurizinho_4902_...1424	Bacurizinho (2008 declared)	133,984 ha	Forest –6.3%, +12,694 ha pasture — tenure gap
Kayapo_23001_...1455	Kayapó	3,283,191 ha	Mining +1,352%, Pasture +1,352% — extractive lockstep
Betania_6201_...1444	Betânia	122,347 ha	Floodable Forest –5.3%, Wetland +223% — hydrology

A 7th folder ([Bacurizinho_4901_...1147](#)) is a duplicate 4901 run kept for completeness. [metadata.json](#) mojibake ([Kayapó](#)) is a UTF-8 export bug in the territory-name field only; it does not affect any numeric output.

Krenak (23401) — coastal Atlantic Forest, the recovery case

Critical juncture 1808 (the "Just War" decree); title restored 1997 (STF).

The headline is *forest recovery on former pasture*. Forest Formation more than doubles, 354 ha (8.3%) → 856 ha (20.0%), **+141%**; Pasture falls 2,879 → 2,189 ha (–24%); Mosaic of Uses gains +220 ha.

Hansen adds the crucial caveat: the satellite era did **not** begin with intact forest. Tree-cover 2000 was already only 56.4% (2,414 ha), and 2001–2023 loss was 189 ha (7.8% of that baseline), concentrated in **2014 (63 ha), 2017 (39 ha), 2015 (32 ha)** — fire-driven pasture stress years in Minas Gerais. Hansen "gain" (a 2000–2012 mask) is near zero, so the 502 ha MapBiomas recovery must be **secondary regrowth** the Hansen gain mask cannot see.

What this refines in the chapter: the "89% deforestation" figure is a *1985 baseline*, not a present trajectory. Post-1988 the direction is slow recovery, concentrated on land that used to be pasture — consistent with leased-land returns after 1980 and judicial restitution in 1997. The 2015 Fundão dam collapse does **not** register as canopy loss here; its only pixel trace is the River class shrinking 30 → 7 ha. Krenak's lesson: *policy recognition is necessary but insufficient once path dependence has already exhausted the resource* — the forest was gone before the satellite record began.

Krenak de Sete Salões (23402) — same direction, larger scale

Same story, bigger polygon: Pasture –1,145 ha (–24%), Mosaic +656 ha (+18%), Forest +436 ha (+6%); marginal new uses appear (Coffee +39 ha, Forest Plantation +27.5 ha). Hansen: tree cover 2000 = 78.4%, 2001–2023 loss ≈ **0 ha**. For practical purposes Sete Salões lost *no* canopy over the satellite era — its dynamic is internal mosaic reorganisation (pasture → mosaic → secondary forest), reflecting its status as first a state park and later an Indigenous reserve.

Bacurizinho — the boundary-effect natural experiment

Refuge after the 1901 Alto Alegre rebellion; original block homologated 1983 (4901); expansion declared 2008 (4902); TRF1 finalisation order 2025.

Running the 1983-homologated block and the 2008-declared expansion together is the methodological payoff: two polygons in the *same* biome and pressure profile that differ only in the **legal age of the boundary**.

4901 (original, 82,515 ha) — demarcated and holding

Forest falls modestly 82,865 → 79,881 ha (–3.6%); Savanna –40%. The telling signal is what *appears* from zero: Pasture (690 ha), Sugar Cane (583 ha), Soybean (341 ha) — the leakage the 1983 demarcation could not fully prevent. Hansen: tree cover 2000 = 99.8%, 2001–2023 loss 5,908 ha (**7.1%**), peaking in the Maranhão fire years 2017 (1,722 ha) and 2014 (1,542 ha).

4902 (expansion, 133,984 ha) — the tenure gap

Same direction, much higher amplitude. Forest –7,041 ha (–6.3%, over twice the absolute loss of 4901); Savanna –43%. **Pasture appears at 12,696 ha** (vs 690 ha in 4901); "Other non-Vegetated Areas" (mining/quarry scars, bare degraded land) jumps 4 → 183 ha (33×). Hansen: tree cover 2000 = 92.3%, 2001–2023 loss 17,062 ha (**13.6%**).

The boundary effect

Indicator	4901 (1983)	4902 (2008)	Differential
Forest Δ 1985–2023	–3.6%	–6.3%	1.75×
Pasture 2023	690 ha	12,696 ha	18.4×
Hansen loss 2001–2023	7.1%	13.6%	1.91×
Hansen 2014+2017 loss	3,264 ha	6,043 ha	1.85×

Three independent metrics agree on a **1.75–1.91× differential** — a clean internal cross-check, and a range that brackets the Pfaff & Robalino (2017) Amazon protection-effect literature. Bacurizinho's lesson: *the same legal category produces different pixel outcomes when tenure clarity differs*. Legal declaration is necessary but not sufficient; what matters is whether demarcation was physically executed and overlapping private (CAR) claims resolved. The 2025 TRF1 order therefore has a quantifiable expected effect: closing the gap would save ≈9,000 ha of Hansen canopy over a 23-year horizon.

Kayapó (23001) — the mining–pasture lockstep

1970s BR-163 contact disaster; Altamira Gathering 1989; ratified 1991.

At 3.28 M ha this is the largest polygon in the thesis, processed in four quadrants (§Methodology 6). Two effects only appear at this scale.

The lockstep. Between 1985 and 2023, Pasture and Mining grow at the *same* relative rate inside the demarcated polygon — both **+1,352.6%**:

Class	1985 (ha)	2023 (ha)	Δ ha	Δ %
Forest Formation	2,875,570	2,821,132	–54,438	–1.9%
Pasture	2,105	30,597	+28,492	+1,352.6%
Mining	951	13,825	+12,874	+1,352.6%
Floodable Forest	170,632	161,178	–9,454	–5.5%
Wetland	3,647	2,049	–1,598	–43.8%

The equal percentage is a numeric coincidence, but the story is real: the two extractive land uses expand **together, not sequentially**, inside a mostly intact forest. The 57 new mining-process filings of 2026 (§9.6) are the next chapter of a process the satellite record can already see.

The Hansen warning. MapBiomas shows Forest Formation –1.9%; Hansen shows a 14.2% canopy-loss footprint over 2001–2023 — because Hansen captures degradation, thinning, and burn-area hits that MapBiomas' persistence logic re-absorbs as the same class. The annual series is post-2017 heavy: **2017 (73,644 ha)**, 2020 (32,074 ha). The **2024 row (305,250 ha)** is a wildfire scar, not canopy removal — the caveat that drove the Collection 10/11 upgrade.

The hydrological footprint. A quieter signal: Wetland collapses 44% (3,647 → 2,049 ha) and River/Lake shrinks 6%, alongside −5.5% Floodable Forest. This coincides with Belo Monte's 2011 licensing and 2016 commercial start even though the dam sits ~900 km downstream — the framework registered a water-regime signal the forest-class series alone would have missed.

Kayapó's lesson: *strong institutional capacity plus international visibility produces the cleanest demonstration of policy effectiveness in the dataset* — but the protective effect now depends on holding back the mining pipeline, and even that strength is not autonomous from national political cycles (the 2019 rollback shows as a 2020 loss uptick).

Betânia (6201) — the hydrological re-classification case

Missionary land purchase in the 1960s; delimited 1985, declared 1991, homologated 1995.

The lowest-deforestation case — but the dominant dynamic is *not* agricultural. Forest Formation barely moves (−0.5%, −341 ha) and Floodable Forest declines only 5.3% (−2,496 ha); meanwhile **Wetland triples (+223%, +2,297 ha)** and Grassland grows ten-fold (+932%, +698 ha):

Class	1985 (ha)	2023 (ha)	Δ ha	Δ %
Forest Formation	63,304	62,963	−341	−0.5%
Floodable Forest	47,318	44,822	−2,496	−5.3%
River, Lake, Ocean	10,106	9,607	−500	−4.9%
Wetland	1,030	3,327	+2,297	+222.8%
Grassland	74	772	+698	+932.0%

This is a **hydrological footprint, not an agricultural one**: Floodable Forest converting to open Wetland and Grassland, paired with River-class shrinkage, is consistent with the Solimões várzea flood-regime change documented for 2010–2020 (extreme low-water years). Hansen: tree cover 2000 = 92.2%, 2001–2023 loss 4,254 ha (**3.7%** — the lowest in the set), with the only >500 ha year being **2023 (789 ha)**, coinciding with the declared health emergency.

What this refines in the chapter: the "COVID/health emergencies are non-deforestation shocks" framing is *half* confirmed — 2020 is quiet (281 ha, at the multi-year median), but **2023 is a real +180% pressure spike**. And the "minimal deforestation (❤️ %)" claim is slightly out of date (Hansen puts it at 3.7%). Betânia's lesson: *where path dependence runs cleanly through Indigenous stewardship and recent recognition, the satellite record shows what an effective policy regime looks like* — and the absence of a pixel transition (COVID) is itself a meaningful finding when the hypothesis predicts stability.

How to open the outputs

Each folder's **figures/* .html** are interactive Plotly charts; the **exports/png_flat/** and **exports/png/trees** hold 2800×1800 PNGs with identical filenames per territory (so ***__transitions_sankey.png** lines up across cases). The single best figure per case study is **transitions_sankey.html** (the 1985 → 2023

flow) paired with `hansen_gfc_loss_by_year.html` (the policy-lag series). The raw numbers above all come from `territory/<name>/comparison_1985_vs_2023.csv` and the two `hansen_gfc_*.csv` files.